Cambridge (CIE) A Level Maths: Mechanics

Your notes

Momentum & Collisions

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Your notes

Momentum

What is momentum?

- Any object that has mass and is moving has **momentum**
- **Momentum** measures the quantity of motion that an object has
- The **momentum** of a particle is defined as the **product** of its **mass ($m$kg)** and its **velocity (v m s^{-1})**
  - Momentum $= mv$
- The SI unit for momentum is kg m s^{-1} (equivalent to newton seconds N s)
- Momentum is a **vector** quantity - so it has a **magnitude** and **direction**
  - The **direction of the momentum** of a particle is the **same** as the **direction of motion** of the particle
  - The **momentum is negative** if the **velocity is negative**
  - If an object is falling vertically then momentum will be downwards

What happens when two objects are in contact?

- If two objects are in contact with each other then by **Newton's Third Law** there will be **equal** and **opposite** reaction forces
- One object gains momentum from the impact and the other loses momentum
- The change in momentum will be equal for each particle provided there are no external forces
  - Remember momentum is a vector and can be negative
- For example, consider the scenario where a tennis ball is struck by a racket, suppose the momentum of the racket and the ball before the impact is 5 kg m s^{-1} and -3 kg m s^{-1} and the momentum of the racket afterwards is -2 kg m s^{-1}
  - This means the racket has a loss of 7 kg m s^{-1} in momentum
  - Therefore the ball has a gain of 7 kg m s^{-1} in momentum
  - Therefore the momentum of ball after the impact is 4 kg m s^{-1}

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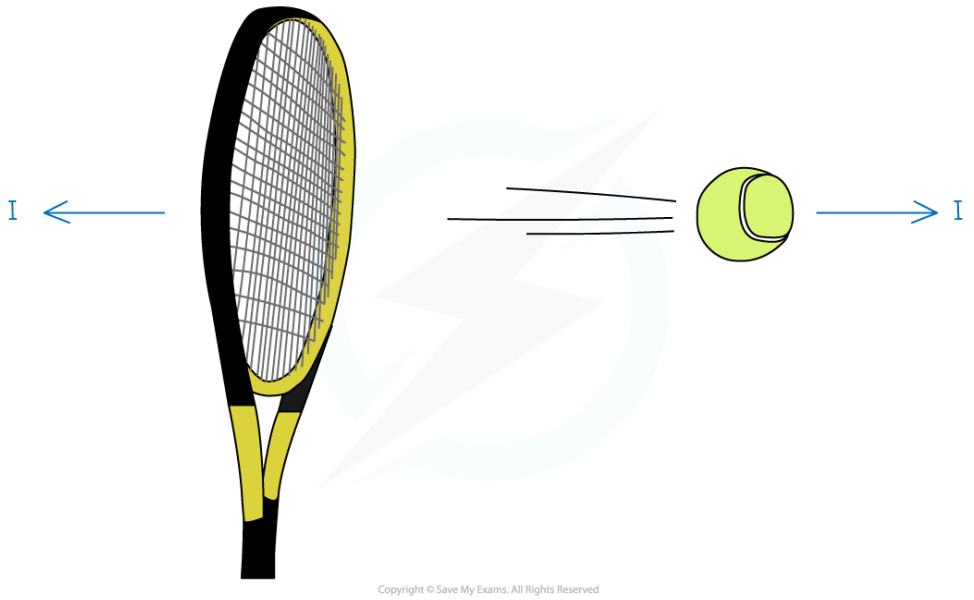

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Worked Example

A dog of mass 15 kg is running with speed $6\ \mathrm{m\ s^{-1}}$.

Find the momentum of the dog.

A sketch can make it clear

Use formula "mv"

Momentum = 15 × 6

90 kg m s⁻¹

Include units kg m s⁻¹

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Collisions

What is a direct collision?

- A **direct collision** is when two objects are travelling along the **same straight line** when they collide
- **Before** the collision:
  - **One** of the objects could be **stationary**
  - The two objects could be travelling in the **same direction** with the faster object behind the slower one
  - The two objects could be travelling in **opposite directions** towards each other
- **After** the collision:
  - **One or both** of the objects could be **stationary**
  - The two objects could be travelling in the **same direction** with the faster object in front of the slower one
  - The two objects could be travelling in **opposite directions** away from each other
  - The two objects could **coalesce** (merge to form one object) and travel in either direction
- **Explosions** work like direct collisions and are when an object **separates** into two objects travelling along the same straight line
  - An example of this is a **bullet being fired from a gun**, the bullet moves forwards and the gun recoils backwards
  - For an explosion it is possible that the object is **initially stationary** and then splits into two objects **moving in opposite direction**

Conservation of Momentum

What is meant by conservation of momentum?

- The **principle of conservation of momentum** states that when two objects collide the **total momentum is unchanged**
  - **Total** momentum **before** collision = **Total** momentum **after** collision
  - This only works if there are **no external forces** acting on the objects
- If an object changes direction after a collision then its velocity changes between positive and negative
  - It is important to be clear about which direction is positive

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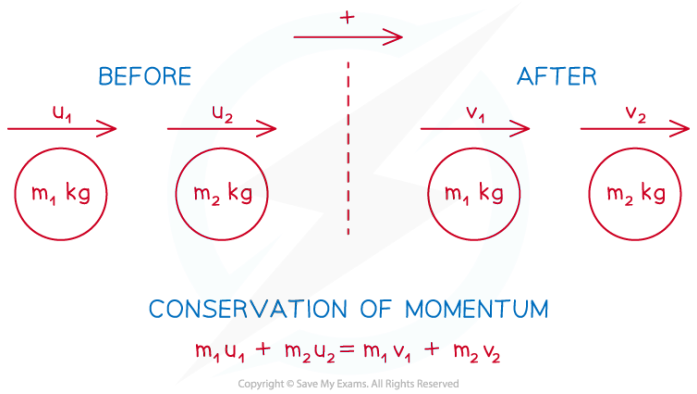
- It can be written as: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$
  - One object has mass m_1 kg, velocity u_1 m s^{-1} before the collision and v_1 m s^{-1} after the collision
  - The other object has mass m_2 kg, velocity u_2 m s^{-1} before the collision and v_2 m s^{-1} after the collision



How do I use conservation of momentum to solve collision problems?

- **STEP 1**: Choose the **positive direction**
- **STEP 2**: Draw a **before/after diagram**
  - Clearly show the mass, speeds and directions
  - If a direction is unknown, then choose any direction and if you get a negative value for its velocity it means it is travelling in the opposite direction
  - If the two objects coalesce then you can either consider them as two particles moving in the same direction with the same speed or consider them as one particle and add together their masses
- **STEP 3**: Form an **equation** using the **conservation of momentum**
  - Be careful with negatives
  - If an arrow is in the opposite direction to the positive direction, then its velocity is negative
- **STEP 4**: **Solve** and give answer in **context**
  - You might need to find the speed and/or direction after a collision

Worked Example

Your notes

Two particles P and Q, with masses 3 kg and 5 kg respectively, are travelling in opposite directions towards each other along the same straight line on a smooth horizontal table when they collide directly. Immediately before the collision the speeds of P and Q are 4 m s^{-1} and 2 m s^{-1} respectively. Immediately after the collision the direction of motion of P is reversed and its speed is 1 m s^{-1}.

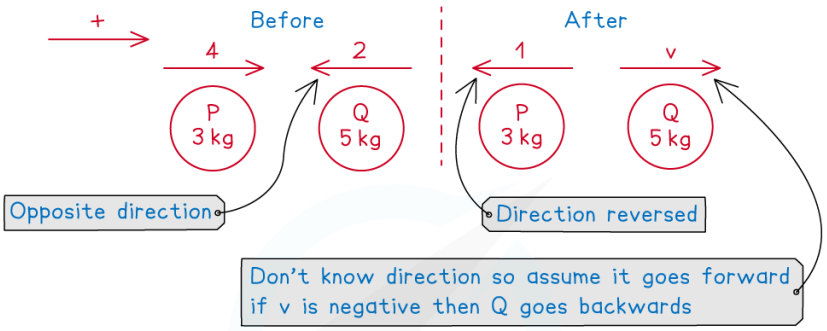
(a) Find the speed of **Q** immediately after the collision.

(b) State whether the direction of motion of **Q** is changed by the collision.

Step 1: Choose positive direction

Let the initial direction of P be positive

Step 2: Draw a before/after diagram



Step 3: Use conservation of momentum

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$3(4) + 5(-2) = 3(-1) + 5(v)$

Negative as arrows go backwards

$12 - 10 = -3 + 5v$

$2 = 5v - 3$

Step 4: Solve

$5v - 3 = 2$

$v = 1$

a) Speed = 1 ms^{-1}

b) $v = 1$ is positive so Q moves forwards after collision

Before collision Q moved backwards

Direction of motion of Q is reversed

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Examiner Tips and Tricks

- Always draw clear diagrams with arrows!
- When considering directions, use common sense - if two particles are travelling in the same direction then they will not collide if the faster one is in front. Two particles can't go through each other (at least not at this level of mathematics...) so if two objects are travelling towards each other, then at least one of them must change direction after the collision.

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Collisions – Multiple Collisions

Can there be multiple collisions?

- After **two objects collide** it is possible that one (or both) of them **collides with something else** such as
  - A **third object**
  - A **wall** (perpendicular to the motion)
- Deal with **each collision separately** and use the steps for direct collisions

Can there be a second collision between the original two objects?

- Let A, B and C be three objects travelling in the **same straight line** and suppose **A and B collide directly** and subsequently **B and C collide directly**
- After the collisions between **A and B** and **B and C** there will be a **second collision** between **A and B if**:
  - **One** is **stationary** and the **other** is **travelling towards** it
  - Both are **travelling in opposite** directions **towards** each other
  - Both are **travelling in the same** direction and the **one in front is slower** than the one behind
- The process is **similar** if object **C is a wall**
  - After **B collides with the wall** its **direction will be reversed** so it will be travelling **towards A**
  - B will **collide with A** again if its **velocity in that direction** is **greater** than the velocity of A in that direction
  - To help you work out the speed of B after hitting the wall you will be given extra information such as the **change in momentum of B** or **loss in kinetic energy**

Worked Example

Three uniform balls A, B and C of equal radius, and mass 0.1 kg, 0.2 kg and 0.3 kg respectively, can move along the same straight line on a smooth horizontal table with B in the middle of A and C. A and B are projected towards each other in opposite directions with speed $10\ \mathrm{m\ s^{-1}}$ and $2\ \mathrm{m\ s^{-1}}$ respectively while C is at rest. A and B collide directly which does not change the direction of motion of A and subsequently A moves with speed $1\ \mathrm{m\ s^{-1}}$.

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(a) Show that the speed of B immediately after it collides with A is $2.5\ \mathrm{m\ s^{-1}}$.

(b) In the subsequent motion, B collides directly with C. Immediately after this collision, C moves with speed $1.5\ \mathrm{m\ s^{-1}}$. Determine if there will be a second collision between A and B.

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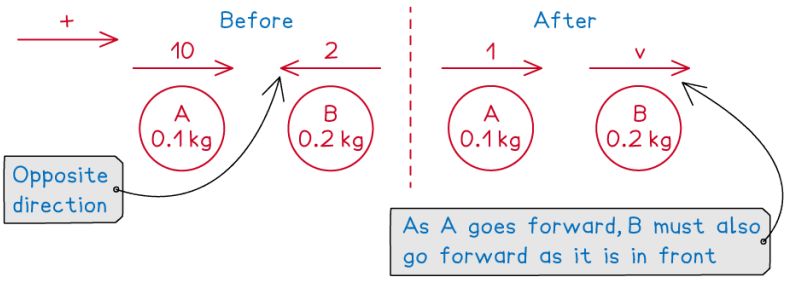
a) Look at the collision between A and B

Step 1: Choose positive direction

Let the initial direction of A be positive

Step 2: Draw a before/after diagram



Step 3: Use conservation of momentum

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$0.1(10) + 0.2(-2) = 0.1(1) + 0.2(v)$

$0.6 = 0.1 + 0.2\,v$

Step 4: Solve

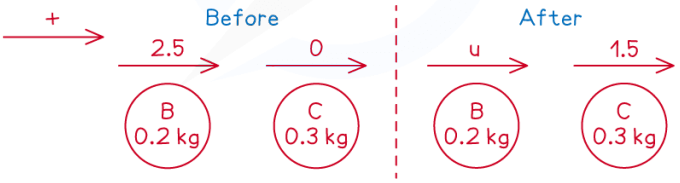
$v = 2.5$

Speed of B = $2.5\ \mathrm{m\ s^{-1}}$

b) Look at the collision between B and C

Step 1: Choose positive direction

Step 2: Draw a before/after diagram



Step 3: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$0.2(2.5) + 0.3(0) = 0.2(u) + 0.3(1.5)$

$0.5 = 0.2u + 0.45$

Step 4: $u = 0.25$

Look at velocities of A and B after collisions

A and B are traveling in the same direction

A's speed is greater than B's speed

There will be another collision between A and B

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Examiner Tips and Tricks

- These questions can be difficult to visualise in your head so draw simple diagrams to show each collision.
- Use common sense, think how many possible (or impossible) ways there are for objects to move after the first collision. You will often have to consider the speed of one or more objects to decide if a second or third collision is possible.
- Questions can ask you to find the loss in kinetic energy so be sure to find the total kinetic energy **before** a collision and compare it to the total kinetic **energy** after a collision.
- These questions can involve lots of algebra, negatives and inequalities so do not rush them as you might make a silly mistake which can affect subsequent parts.

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